TEXORA · LECTURE SHEET

Water and Its Softening

The largest raw material of wet processing: where process water comes from, the quality limits it must meet, what hardness is and how it is measured and calculated, the damage hard water does, the softening treatments from lime–soda to ion exchange, re-use and rainwater harvesting, and the ZDHC rules on hazardous chemicals.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 2 · LEVEL 2 TERM 2

1. Why water dominates wet processing

Every preparatory, dyeing and washing operation is carried out in water. A wet-processing plant typically uses **50–150 litres of water per kilogram of fabric** across all steps, so water is simultaneously the solvent, the transport medium, the heating medium and the largest single input by weight. Its quality decides chemical consumption, shade reproducibility and machine maintenance; its conservation decides whether the plant can run at all in water-stressed regions.

2. Sources of water

- **Surface water** (rivers, canals, ponds): abundant but variable — turbidity, dissolved organic matter, seasonal hardness changes; needs clarification and filtration.
- **Groundwater** (tube wells): the main source for most Bangladeshi mills — clear but often high in hardness, iron and manganese; levels are falling, which raises both cost and regulatory pressure.
- **Municipal supply:** treated and consistent, but costly at industrial volumes.
- **Rainwater:** naturally soft (near-zero hardness), ideal for processing, but seasonal — useful only with harvesting and storage.
- **Recycled process water:** treated effluent or recovered condensate/rinse water returned to selected steps.

3. Quality requirements for process water

PARAMETER	DESIRED LIMIT (GUIDE)	WHY IT MATTERS
Hardness (as $CaCO_3$)	< 50 mg/L for dyeing; near 0 for final rinses	Prevents soap waste, deposits, unlevel dyeing

pH	6.5–7.5	Stable bath chemistry; no fibre attack
Iron (Fe)	< 0.1 mg/L	Iron catalyses peroxide decomposition (pinholes), causes rust spots and dull shades
Manganese	< 0.05 mg/L	Brown staining; catalytic bleach damage
Turbidity / suspended solids	Clear, < 5 NTU	Deposits on fabric, specks, machine fouling
Total dissolved solids	Low and constant	Salt load affects dye exhaustion and reproducibility

4. Hardness: types, units and calculation

Hardness is the concentration of dissolved calcium and magnesium salts, expressed as an equivalent amount of $CaCO_3$ (mg/L = ppm). It has two components:

- **Temporary (carbonate) hardness:** $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ — removed by boiling, which precipitates $CaCO_3$.
- **Permanent (non-carbonate) hardness:** chlorides, sulphates and nitrates of Ca^{2+} and Mg^{2+} — not removed by boiling.

$$Ca(HCO_3)_2 \xrightarrow{\text{(heat)}} CaCO_3\downarrow + CO_2\uparrow + H_2O$$

Units: 1 °Clark (English) = 14.3 mg/L; 1 French degree = 10 mg/L; 1 German degree (°dH) = 17.8 mg/L as $CaCO_3$. Water below ≈ 60 mg/L is called soft, 60–120 moderately hard, 120–180 hard, above 180 very hard.

Converting an ion concentration to hardness as $CaCO_3$: multiply by the ratio of equivalent weights, Eq. wt. $CaCO_3$ = 50.

$$\text{Hardness contribution (mg/L as } CaCO_3) = (\text{mg/L of ion}) \times 50 \div (\text{equivalent weight of the ion})$$

WORKED EXAMPLE

A bore-well sample contains Ca^{2+} 40 mg/L and Mg^{2+} 12 mg/L. Equivalent weights: Ca^{2+} = 40/2 = 20; Mg^{2+} = 24.3/2 ≈ 12.15.

From calcium: 40 × 50/20 = 100 mg/L as $CaCO_3$. From magnesium: 12 × 50/12.15 ≈ 49 mg/L as $CaCO_3$.

> **Total hardness ≈ 149 mg/L as $CaCO_3$** — classified as hard, and unsuitable for dyeing without softening.

5. Consequences of hard water in wet processing

- **Soap waste:** Ca^{2+}/Mg^{2+} precipitate soap as insoluble lime soap (curd) — detergent is consumed before it can clean, and the sticky curd deposits on fabric and machine walls.
- **Unlevel dyeing:** metal ions complex with many dyes, shifting shade and reducing solubility; deposits cause resist spots.
- **Bleaching damage:** Fe^{3+}/Cu^{2+}/Mn^{2+} (often accompanying hard water) catalyse violent peroxide decomposition, producing local tendering and pinholes.
- **Boiler and pipe scale:** $CaCO_3$/$CaSO_4$ scale insulates heat-transfer surfaces — fuel waste and tube failure.
- **Handle and quality:** lime-soap deposits give fabric a harsh, grey, uneven handle and attract soil.

6. Softening treatments

6.1 Lime–soda process

Lime $Ca(OH)_2$ removes temporary hardness and magnesium; soda ash Na_2CO_3 removes permanent calcium hardness:

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2\,CaCO_3\downarrow + 2\,H_2O$$

$$MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCl_2 \quad ; \quad CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2\,NaCl$$

The precipitates settle as sludge. Cheap for large volumes, but leaves residual hardness of about 30–50 mg/L and needs careful dosing and sludge handling.

6.2 Zeolite (permutit / base-exchange) process

Water passes through a bed of sodium zeolite (today usually a synthetic ion-exchange resin in Na^+ form). The resin exchanges its Na^+ for Ca^{2+} and Mg^{2+}:

$$Na_2\text{–Resin} + Ca^{2+} \rightarrow Ca\text{–Resin} + 2\,Na^+$$

When exhausted, the bed is regenerated with brine (NaCl). Output hardness can approach 0–5 mg/L, but total dissolved solids are unchanged (ions are swapped, not removed) and iron fouls the resin.

6.3 Ion-exchange demineralisation

Two beds in series — a cation exchanger in H^+ form and an anion exchanger in OH^- form — remove *all* dissolved salts, giving near-pure water for boilers and critical processes. Regeneration uses acid and caustic soda. Mixed-bed units polish to the lowest conductivities.

6.4 Sequestering agents (internal softening)

Where plant softening is impractical, chelating agents (EDTA, DTPA, phosphonates, polyacrylates) are dosed into the bath: they lock Ca^{2+}/Mg^{2+}/Fe^{3+} into soluble complexes so the ions cannot precipitate soap or catalyse bleach damage. The hardness is masked, not removed — dosing must match the ion load.

METHOD	RESIDUAL HARDNESS	REMOVES TDS?	BEST USE
Boiling	Temporary part only	No	Small scale, laboratory
Lime–soda	≈ 30–50 mg/L	Partly (precipitation)	Large-volume first stage
Zeolite / Na-resin	≈ 0–5 mg/L	No (ion swap)	Process water, boilers (low pressure)
Demineralisation	≈ 0, all salts removed	Yes	High-pressure boilers, critical dyeing
Sequestering agents	Masked in-bath	No	In-process protection, bleach baths

7. Consumption, re-use and recycling

Consumption is cut first by machine choice (low-liquor jets and continuous ranges use far less than winch or overflow machines), then by practice: counter-current washing, reusing cooling water and condensate, and combining steps (one-bath scour–bleach). **Re-use** returns relatively clean streams — final rinse water, cooling water — to washing steps. **Recycling** treats effluent (through an ETP with biological and, increasingly, membrane stages) back to process quality; reverse-osmosis polishing can recover 60–80% of treated effluent for re-use. Every litre recycled is a litre not drawn from a falling aquifer — in Bangladesh this is now a compliance as well as a cost issue.

8. Water conservation: rainwater harvesting

Rainwater is the softest water available to a mill (hardness near zero, no iron), and the monsoon delivers it in quantity. A factory system collects roof run-off, filters out leaves and dust, stores it in tanks or a lined pond, and feeds it to process use — often blended with groundwater to cut overall hardness and pumping.

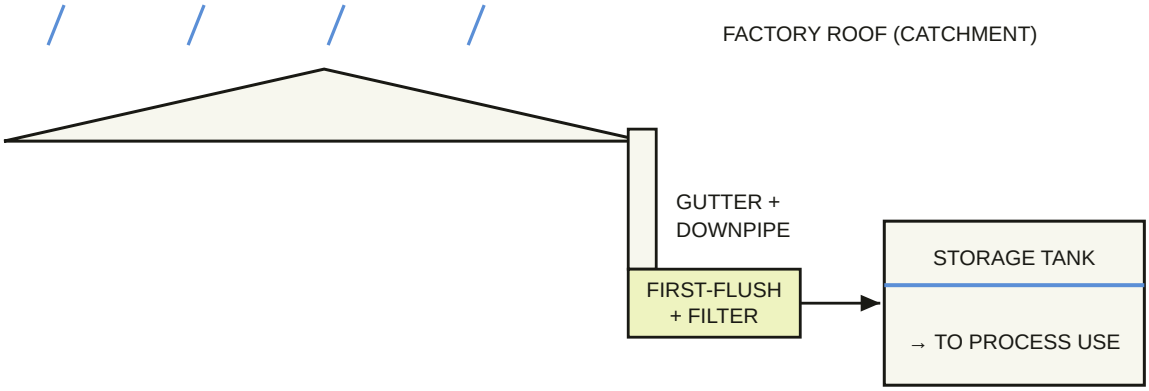


Fig. 2 — Roof rainwater harvesting: catchment → gutter → first-flush/filter → storage → process

Fig. 2 — A mill rainwater-harvesting train. First-flush diverters reject the dirty first minutes of a storm.

9. ZDHC and banned chemicals

ZDHC (Zero Discharge of Hazardous Chemicals) is an industry programme whose **MRSL** (Manufacturing Restricted Substances List) bans or limits hazardous chemicals in production inputs. A mill's water story is incomplete without it: banned substances reach water through chemical use, so controlling the chemical inventory is part of water stewardship. Groups restricted/banned under ZDHC MRSL and brand RSLs include:

- **Azo dyes** that can cleave to release any of the listed carcinogenic aromatic amines.
- **APEO** (alkylphenol ethoxylates) — endocrine-disrupting surfactants, replaced by fatty-alcohol ethoxylates.
- **Formaldehyde** above limits (easy-care finishes, pigment printing binders).
- **Heavy metals** — chromium(VI), lead, cadmium, mercury in dyes, pigments and mordants.
- **Chlorinated phenols** (PCP, TeCP), chlorobenzenes, and short-chain chlorinated paraffins.
- **Phthalates** in plastisol prints and coatings; **organotin** compounds; PFC/PFAS water-repellent finishes (increasingly phased out).

Compliance is shown by buying chemicals with MRSL conformance (e.g. ZDHC Gateway levels), testing wastewater, and substituting safer alternatives — the same discipline that keeps banned chemistry out of the effluent keeps it off the fabric.

10. Glossary — key terms

- **Hardness:** Ca^{2+} + Mg^{2+} content expressed as mg/L $CaCO_3$ equivalent.
- **Temporary / permanent hardness:** bicarbonate hardness removable by boiling, versus chloride/sulphate hardness that is not.
- **Sequestering agent:** a chelating chemical that binds metal ions into soluble, inactive complexes.

- **MRSL:** Manufacturing Restricted Substances List — input-chemical restrictions under ZDHC.
- **ETP:** effluent treatment plant.

11. Key points

- Process water must be soft (< 50 mg/L as $CaCO_3$), near-neutral, and virtually free of iron and manganese.
- Hardness is calculated by converting each Ca/Mg ion to its $CaCO_3$ equivalent via equivalent weights.
- Hard water wastes soap, causes unlevel dyeing and scale, and its iron content destroys bleached fabric.
- Choose softening by target quality: lime–soda for bulk, ion exchange for near-zero hardness, sequestrants for in-bath protection.
- Re-use, recycling and rainwater harvesting reduce both cost and groundwater dependence; ZDHC MRSL controls the chemicals entering the water.

12. Summary

Water is the working medium of all wet processing, and its hardness — the calcium and magnesium it carries — is the main quality risk. Hardness is measured as $CaCO_3$ equivalent and removed by precipitation (lime–soda), exchange (zeolite/resin, demineralisation) or masked by sequestering agents. Conservation closes the loop: low-liquor machines, re-use of clean streams, ETP recycling and rainwater harvesting cut consumption, while the ZDHC programme keeps hazardous chemicals out of the water in the first place.

13. Practice

Answer in your own words; brief model answers follow.

1. A water sample contains Ca^{2+} 60 mg/L and Mg^{2+} 24 mg/L. Calculate the total hardness as $CaCO_3$ and classify the water.
2. Why does boiling remove only temporary hardness? Write the reaction involved.
3. A bleach house reports occasional pinholes in peroxide-bleached fabric. The water is soft but contains 0.4 mg/L iron. Explain the connection and give two remedies.
4. Distinguish zeolite softening from demineralisation in terms of what leaves the water.
5. Name four chemical groups restricted under the ZDHC MRSL and state why each is restricted.

Answer key

1. Ca: 60 × 50/20 = 150 mg/L. Mg: 24 × 50/12.15 ≈ 99 mg/L. Total ≈ 249 mg/L as $CaCO_3$ — very hard; softening is essential before dyeing.
2. Boiling decomposes the soluble bicarbonates to insoluble carbonate, which precipitates: $Ca(HCO_3)_2 \rightarrow CaCO_3\downarrow + CO_2 + H_2O$. Chlorides and sulphates of Ca/Mg are thermally stable in solution, so permanent hardness stays.
3. Fe^{3+} catalyses rapid local decomposition of H_2O_2, generating aggressive radicals that attack cellulose at points — local tendering appears as pinholes. Remedies: remove iron at source (aeration/filtration or ion exchange) and add a sequestering agent/stabiliser system to the bleach bath to bind residual iron.
4. Zeolite swaps Ca^{2+}/Mg^{2+} for Na^{+}: hardness ions leave but are replaced, so total dissolved solids stay roughly constant. Demineralisation removes cations *and* anions (replaced by H^{+} and OH^{-} that combine to water), so TDS falls to near zero.
5. Any four, e.g.: azo dyes releasing banned amines (carcinogenic); APEOs (endocrine disruptors, persistent); chromium(VI) and other heavy metals (toxic, carcinogenic); formaldehyde above limits (sensitiser/carcinogen concern); chlorinated phenols (toxic preservatives); phthalates (reproductive toxins).
